

6(a) (i) E.m.f. is the energy converted into electrical energy from other forms of energy per unit charge in driving a charge round a complete circuit.

Potential difference across XY the energy converted from electrical energy to other forms of energy per unit charge passing across the two points.

B2

(ii)

$$V_X = \frac{E}{2} \quad V_Y = \frac{R}{R_T + R} E$$

$$V_{XY} = V_X - V_Y = \frac{E(R_T + R) - 2RE}{2(R_T + R)} = \frac{E(R_T - R)}{2(R_T + R)}$$

M2

(b) (i) By decreasing the resistance of the variable resistor, the p.d. across the slide-wire increases.

B1

This increases the potential gradient of the slide-wire, and hence a shorter L_Y is required to achieve the same p.d. as before.

B1

(ii) Ratio $L_Y : L_Z$
 $= V_{XY} / V_{XZ}$
 $= 3R / 5R$
 $= 0.6$

C1

A1

(iii) By potential divider principle, the ratio of voltage across V_{XY} and V_{XZ} remains unchanged.

M1

Hence, no change to ratio $L_Y : L_Z$.

A1

DO NOT WRITE IN THIS MARGIN

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7 (a) When the switch is closed, current flows downwards in the wire. The magnetic B1